Systematic Literature Review and Comparative Performance Analysis of SQL and NoSQL Databases in Big Data Applications

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***Abstract*— Today, in the digital moment, when the presence of the Internet in daily life is all around, management of data has become an indispensable part in any kind of organization irrespective of sector. Large-scale data applications are remarkably influenced in prosperity by the choice between SQL and NoSQL databases. This research deals with issues regarding which among database types- SQL or NoSQL-is more effective in Big Data environments. The paper consequently benchmarks the key performance indicators such as storage capacity, query execution time, and scalability through a systematic review of literature. SQL databases have been found to be robust with respect to structured data and complex queries, maintaining transactional integrity and consistency. They are, however, limited in scalability and flexibility when handling unstructured data, thus less suitable for specific big data applications. On the other hand, NoSQL databases support availability and flexibility quite well, handle unstructured data in very large volumes, and give support for a variety of data models. This study concludes that organizations have to align database choice with the type of data an organization deals with and scalability requirements for optimal performance in Big Data applications.**

Keywords—SQL databases, NoSQL databases, Big Data applications, performance analysis, database optimization

I. Introduction

The transformation into digital has mainly changed how information is used and controlled in companies, thereby emphasizing the importance of information in various areas. This is because more and more data of all sorts is being generated at a rate that cannot be expressed in simple numbers. The necessity for efficient and expanding database solutions could not have been better underscored as more information grows by large magnitudes daily in terms of amount, rate as well as type [1][2][3].

Structured Query Language databases have always been known as the choice for storing data due to their strength when it comes to managing structured data having been for long. Developed initially by IBM in 1970 as SEQUEL, SQL has grown over time and is now what we use for our relational databases as it helps in getting this kind of information by use of structured queries [4]. Big Data applications have changed everything by bringing varied types of data that do not have a uniform shape leading us to notice that there are some problems like scale and flexibility which are part of SQL-based databases [1].

NoSQL databases, including MongoDB and CouchDB have come up as the most competitive alternative in addressing such challenges offering data models that are more adaptable handling large amounts of non-related and semi-structured data. Traditional SQL databases are based on fixed schemas and tabular arrangements unlike NoSQL which is flexible with its schema design allowing for easier management of information across distributed systems and gives room for more rapid growth [1]. NoSQL technology rise results primarily from web 2.0 companies need who give priority to dealing with user-generated materials, they would like it be easy for them, also they want their systems to blend smoothly with those of their clients thus propelling this development forward [3]. Every NoSQL database type which includes key-value stores, document stores as well as column family stores, and graph databases respectively offer unique benefits that based on the kind of applications one may have as well as handling such information [2].

Besides, a variety of NoSQL databases usually make them mixed model based and this further complicates transferring information across various cloud storage services (CSPs) [2]. It is for this reason that there is a necessity for strong data integration strategies which can enable us use different NoSQL solutions optimally. This is why high-speed databases have become a key factor in determining whether an application will prosper in today’s world dominated by digital revolution that requires supporting many users at once. Speedy databases not only make it possible to process or analyze large streams of information right away but also scale effortlessly with time to meet increased needs [2].

The objective of this paper is to conduct a comprehensive analysis and optimization of the performance between NoSQL and SQL databases in Big Data applications. By benchmarking key performance indicators such as data storage, query execution time, and scalability, we aim to identify the strengths and weaknesses of each database type. The remainder of the paper is arranged as follows: first we discuss the methodology of our study and then is followed by the result of systematic literature review (SLR), in section 2. In section 3, we present a detailed comparison of NoSQL and SQL databases based on performance metrics. Finally, we conclude our work with suggestions for the next research in section 4.

II. Research Methodology

A. Planning the Review

This study looks at respected scholarly articles that have a link to certain keywords in the title or content. Papers, journals, and theses were sourced from a variety of electronic databases. These electronic databases are certified, and they only contain academic documents that have been accepted internationally.

TABLE I. SEARCH TERMS & LOCATIONS

Search locations	Search terms
IEEE Xplore Google Scholar Science Direct	("SQL" AND "NoSQL performance") OR "SQL performance" OR "NoSQL performance" "SQL database" OR "NoSQL database" OR "SQL" OR "NoSQL" OR "SQL in big data" OR "NoSQL in big data"

To ensure the eligibility of the works selected, we apply the selection criteria in Table II using search terms and search locations in Table I.

TABLE II. SELECTION CRITERIA

Inclusion Criteria	Exclusion Criteria
Literature publications that are less than five years old or over 2019. Studies that focus on the performance analysis of SQL and NoSQL databases in big data applications. Research that reports the performance of SQL or NoSQL databases. Research papers or journals that are open access. Literature consists of credible studies that involve rigorous testing and provide verifiable material.	Publication of literature that is more than five years old or before 2019. Studies that do not discuss the performance analysis of SQL and NoSQL databases in big data applications. Research papers or journals that are not open access.

Several research questions are formulated in order to assess all of the publications and extract important information. These research questions, which aim to make data collection and analysis easier, are:

- RQ1: What are the strengths and weaknesses of SQL and NoSQL in Big Data applications?
- RQ2: How do the performances of SQL and NoSQL compare in terms of speed, scalability, and storage efficiency in Big Data applications?
- RQ3: In what context is SQL superior to NoSQL or vice versa in Big Data applications?

B. Conducting the Review

The methodology used for this study is a Systematic Literature Review (SLR). We collected some data from journals, proceedings, and literature reviews related to the 3 research questions we have designed.

When searching for previous studies using search terms and locations in Table I, we have found 2.399 documents were identified through database searching, with an additional 114 records identified from other sources, bringing the total to 2.513. After removing duplicates, 2.134 records remained. These records were then screened based on their titles and abstracts, resulting in the exclusion of 1.928 records. Consequently, 1.018 records were assessed for eligibility through a full-text review. Then, we filter all documents we found using selection criteria defined in Table II. As a result, we found 30 previous studies eligible to be analyzed for this study as they all met the selection criteria.

We used PRISMA checklist methodology to help us evaluate the papers used in this study. More detail of this process is shown in the PRISMA Flowchart below (Fig. 1).

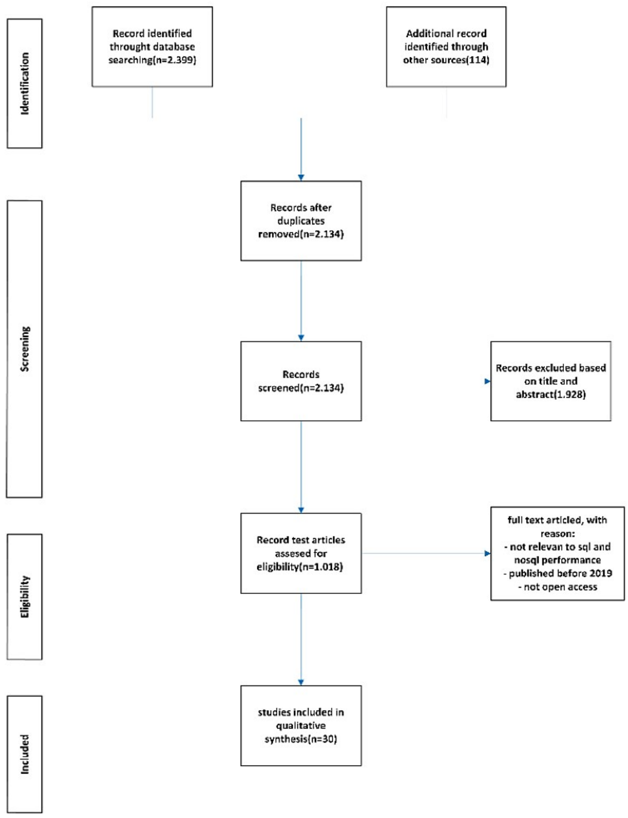

Fig. 1. PRISMA Flowchart

III. RESULT AND DISCUSSION

A. RQ1: What are the strengths and weaknesses of SQL and NoSQL in Big Data applications?

TABLE III. STRENGTHS & WEAKNESSES

Aspect	SQL	NoSQL
Strength	Structured data with predefined schema [5]	Flexible, schemeless design, allowing handling of unstructured or semi-structured data [6]
	Strong transactional support (ACID) [5]	High horizontal scalability across multiple nodes [7]
	Effective for complex queries, joins, and indexing [8]	High performance for read/write operation with large-scale data [9]
	Mature ecosystem with extensive tools and community support [11]	Operational simplicity with features like automatic rebalancing and scalability by adding nodes (e.g., Cassandra) [10]
Weakness	Limited scalability, particularly in terms of horizontal scaling [12]	Lack of standardized query language, leading to potential learning curve and vendor lock-in [13]

	Rigid schema, requiring extensive planning for changes [14]	Often sacrifices ACID properties for scalability and performance (eventual consistency models) [15]
	Potential performance bottlenecks with large data volumes [16]	Generally less mature with fewer tools and community resources [17]
	Complex and labor-intensive to manage 18]	More cost-effective at scale, reducing operational overhead and infrastructure costs [19]

SQL databases effectively manage structured data via predetermined structures, ACID properties ensure the reliability and consistency of information within them, which renders these kinds of applications needing a great deal of support on transactions quite stable. They operate within this goal alongside an established set of tools and community support structures while also allowing for the completion of complex queries, joins as well as index creation. But the downside to using SQL based systems is their inability scale horizontally to accommodate large-scale information hence limited possibilities handling sub-unstructured facts. In addition, SQL based systems tend to slow down when dealing with huge write operations due to excessive data redundancy [20].

NoSQL databases just like Apache Cassandra are developed to be highly scalable and available, hence the best-designed type database for big-data applications [4]. With them one can have an unstructured or semi-structured data storage system without schemas since they have schemeless flexible designs [20]. Thanks to its distributed nature and basic operations, for instance Cassandra's "ring" concept and use of masterless replication means that you can never imagine having slow read/write operations using NoSQL databases [4]. However, lack of a common query language may carve and vendor lock-in and steep learning curves on NoSQL platforms since there is none thus these types often lack standardization techniques. Despite sacrificing ACID properties there are instances where scalability must be maintained because not all programs need this feature. Besides being younger SQL, NoSQL community contains less resources and tools that facilitate its growth such as NoSQL databases whose consistency is eventually achieved through prioritization of availability as well as partition tolerance, and not consistency itself [20].

Apache Cassandra is more acceptable for large data applications because it is scalable, flexible, and more highly performing than SQL databases (NoSQL database). In addition, operationally, simpler and cost-efficient services give a significant advantage this type of database has over SQL systems while undertaking real time analytics of large data sets [4]. In terms of transactional guarantees as well as query complexity; hence differentiating whether one should use either service is on a case-to-case basis [20].

B. *RQ2: How do the performances of SQL and NoSQL compare in terms of speed, scalability, and storage efficiency in Big Data applications?*

TABLE IV. PERFORMANCE COMPARISON

Aspect	SQL	NoSQL
Speed	Good reading performance, easy to code. Slower query runtime and higher latency [21]	High write scalability, efficient for large data, lower latency [22]
Scalability	Vertical scaling, limited for large datasets [23]	Horizontal scaling, handles large unstructured data well [24]
Storage efficiency	Efficient for structured data, issues with redundancy in horizontal scaling [25]	Efficient load balancing, potential data redundancies, uses less CPU and storage [26]

Speed: SQL databases such as MariaDB and Microsoft SQL Server demonstrate good read scalability while being straightforward to comprehend and code. SQL databases perform well for complex queries and joins but might experience increased latencies as well as longer-running operations for huge data capacities [5][19][3]. NoSQL databases (such as MongoDB and Cassandra) lead the pack in terms of write scalability and concurrent user access. These systems effectively support real-time data and very large database sizes regarding speed of operations and time lag features as opposed to SQL databases specifically in cases involving unstructured data [5][19][3].

Scalability: SQL databases depend mainly on vertical scaling and may end up being ineffective when dealing with massive datasets. Horizontal scaling capabilities available in some SQL databases do not perform as well as those provided by NoSQL databases [9][19][3]. NoSQL databases are developed to be scalable horizontally, thus are suitable for managing huge unstructured data. This is achieved through techniques such as sharding which helps distribute data in more than one node ensuring that there is always a node accessible in case one goes down [5][9][19][3].

Storage efficiency: Data redundancy and storage inefficiency are problems they encounter when scaled horizontally but work perfectly when the amount is less. For instances where there should be a high level of data consistency during isolated transactions, SQL databases offer an excellent choice thanks to their use of a relational storage model. The NoSQL models created expressly for fast and efficient handling unstructured big data are MongoDB and Cassandra. They have a document-based schema system with some redundancy while being geared towards load balancing as well as increased fault tolerance. Moreover, in terms of resource utilization that involves CPU load together with available memory disk I/O (input output), though in different directions - lesser is used by MongoDB compared to Cassandra [5][19].

C. *RQ3: In what context is SQL superior to NoSQL or vice versa in Big Data applications?*

SQL Superiority: Structured Data and Complex Queries: SQL databases like MySQL and Microsoft SQL Server are best for situations where data has a clearly defined format and there are some very complex relations. The conditions under which they operate include occasions when maintaining control over the data at any cost is critical, when there is a need for compliance with ACID principles, or regarding intense reliability, like in banking industries [21] [5][9].

Transaction Processing and ACID Compliance: For transactions requiring guarantee and ensuring data integrity that is strict for instance banking applications. ACID transactions have reliable support from SQL databases, thereby being necessary to preserve consistency [21][9]. Normalization and Data Integrity: SQL databases are well-suited for scenarios where data needs to be normalized across multiple tables, ensuring efficient data organization and minimizing redundancy [8][6].

NoSQL Superiority: Scalability and Performance with Unstructured Data: In dealing with large volumes of unstructured or semi-structured data, environments appreciate bright NoSQL databases like MongoDB or Cassandra since they have horizontal scalability that is distributed across nodes and is perfect for applications that need high throughput coupled with very low latency-even at times when working with real-time analytics or IoT for example [4][14][13].

Flexible Data Models and Schema-less Design: NoSQL databases provide flexibility in schema design, allowing for agile development and accommodating diverse and evolving data formats without the need for predefined schemas. This flexibility is advantageous in dynamic environments where data structures can change frequently [9][14]. Big Data Analytics and Flexibility: MongoDB and similar NoSQL solutions excel in handling big data analytics where data variety and volume are significant, enabling efficient storage and retrieval of diverse data types [14][13].

Structured environments that require data integrity and complex queries use SQL databases [27][28]. At the same time, NoSQL databases are better since they can scale up with greater ease or adapt themselves to any schema configurations due to their superior speed in handling unstructured data, hence making them most suitable for today's consistent applications such as internet of things and fast analytics [29][30]. Choosing between SQL and NoSQL depends on the specific needs of the application. If an application requires maintaining structured data with strong integrity and handling complex queries, SQL is the preferred option. Conversely, if the application demands scalable, flexible, and high-performance data management, NoSQL is more suitable.

IV. Conclusion

The study assessed, based on the review of related literature, the performance of SQL and NoSQL databases concerning big data applications using key performance indicators such as speed, scalability, and efficient storage. Our findings thus show that SQL databases are effective in applications where there is the need to handle structured data with complicated queries and high transaction support. They ensure robust data integrity and consistency through ACID properties. Everything is all right, but they create scalability and flexibility issues in the case of unstructured data. These NoSQL databases come with very good scalability, flexibility, and performance to handle large volumes of unstructured data. Hence, they will be suitable for applications that require high speed and low latency, such as IoT platforms and real-time analytics. NoSQL databases are designed to allow flexible schema design, operational simplicity, and cost efficiency. The choice between SQL and NoSQL would thus be informed by the needs of the application at hand in terms of data structure, scalability requirements, and performance metrics.

Further research in this area should shift toward hybrid databases that take advantage of the strengths of both SQL and NoSQL toward addressing a wider range of data management-related challenges. Other interesting dimensions of research may relate to how such emerging technologies as machine learning/artificial intelligence influences database performance and optimization. Case studies in different sectors will also help to validate such findings and provide useful guidelines on how SQL and NoSQL databases can be effectively implemented.

Author's Contribution

The study was conceived and designed by Muhammad Tsaqif Samarta. The systematic literature study are performed by Muhammad Tsaqif Samarta. The writing is guided by Alexander A S Gunawan and M Edo Syahputra. All authors read and approved the manuscript.

Availability Data and Materials

This study utilized publicly available literature from IEEE Xplore, Google Scholar, and Science Direct. These literatures were specifically selected to ensure comprehensive coverage of the relevant research.

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